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Illustrating what a randomized controlled trials is: An incentive Experiment (based on https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0240291)

This text uses terminology presented in Bergenholtz 2025a, which is recommended to be read first.

The Objective: This text presents an experiment that was carried out in the COBE Lab (here at Aarhus University) in 2019/2020. The central aim of this text is to help explain and illustrate what a randomized controlled trial (RCT) is: with a particular focus on the external validity. It's also about recognizing the many decisions that shape an RCT's design and acknowledging how variations in these decisions might lead to different outcomes.

The Research Question: Incentives and performance: In order to understand the nature of this RCT, I'll briefly introduce what the aim of the study was. We investigated if individuals would perform better or worse, when receiving a higher (vs. a smaller) incentive. Some prior studies had shown that if you offer higher incentives (compared to lower incentives, i.e. less money) individuals perform better, while other prior studies showed conflicting results; either showing no effect, or the opposite effect – that people might perform worse if offered a higher incentive (compared to lower). Simply put, economic theory argues that higher incentives lead to higher effort and thus higher performance. In contrast, one can use psychology theory to argue that people are mainly driven by how interesting a task is, and that if the incentives are too high, people might choke under pressure – thus implying that higher (vs. lower incentives) lead to the same or lower performance. Since theory was ambiguous, we generated two opposing hypotheses:

- *Hypothesis 1*: A larger piece-rate reward will lead to better performance.
- *Hypothesis 2*: A larger piece-rate reward will lead to worse performance.

The paper explains this in much more detail, but these are not important for the purpose of this course and topic. The RCT now sets out to test the relation.

Controlled environment: In our controlled social science lab, all participants face the same environment and are to solve identical tasks; a math addition and anagram task (see illustrations at the end of the text). Except for the treatment, all other factors are set up in the same way; the control and treatment conditions took place in the same room, at the same time of the day, with the same tools, the same time span, the same kind of introduction, the same research assistant etc.

Treatment Participants either end up in the treatment or control condition, of course not knowing if they are in one or the other. Some individuals received a higher incentive (5 DKK pr. solved task) vs. a lower incentive (1 DKK pr. solved task). If the results then show an average difference in performance between individuals in the treatment vs. control group, then this should only be due to the difference in incentives (since everything else was the same). Note we can only show if higher is different from lower, not if it is different from giving no incentives - since this was not a condition. Participants in the COBE Lab (rightly so) always have to be offered some money to participate.

Given logistical constraints, we implemented session-level randomization. Thus, every individual in a given session was randomly assigned to receive one particular kind of treatment – rather than randomizing at the individual level. Nevertheless, this still meant that individuals could not self-select into the treatment or control condition, because they did not know what kind of session would be run on a particular day. Furthermore, all sessions were run at the same time of the day, to ensure consistency. Thus, difference in behavior could not be due to either control or treatment primarily being in the morning or in the afternoon. E.g., if our system had led to a situation where treatments primarily were run at 9 am and control primarily at 2 pm, we would have risked systematic differences; maybe certain people show up at 9 am, or maybe behavior is different at 9 am vs 2 pm.

Obviously one also has to make sure that people are informed about the incentives before they perform the tasks. This ensures the direction of causality; incentives → performance.

Internal Validity Considerations: This experiment followed the standards of a RCT, ensuring strong internal validity. Participants did not self-select into a condition, and we randomized people into the different groups to ensure that people in one condition should be as similar as possible to those in the other condition, that is all individuals experience the same, except for the treatment (vs. control). Our randomized design ensures that no confounding variables can interfere, allowing us to observe only the differences created by the varied incentive levels. We therefore don't need control variables as such.

Nevertheless, we still statistically analyze if we there are differences (e.g. in demographics) across the conditions, just to ensure that the randomization actually divided individuals equally across conditions. For example, we can compare the average age of participants to ensure that they were in fact randomly distributed. Just as one sometimes flips a coin to the same outcome several times in a row, random distribution can lead to uneven outcomes, in smaller samples (see Bergenholtz 2025a). The larger the sample size, the less risk of the samples in the two conditions being uneven (that is not identical).

Note that an experiment provides us with average difference. Maybe some individuals are impacted more than others – a direct test comparing conditions doesn't reveal that.

One of the key risks that is difficult to eliminate, is related to such experiments taking place over multiple days and weeks. In principle participants could have heard about the setup before showing up, putting internal validity at risk – again, internal validity relies on people not knowing what conditions others are in, since that knowledge might influence their behavior. We have to rely on the integrity of participants here.

External Validity Considerations: While accepting that the findings hold for this particular sample, in this particular year, with these particular tasks (internal validity), how can we generalize? The following criteria are based on those presented in Bergenholtz 2025a). They are supposed to assist the reader of an experiment to provide a detailed assessment of the potential to generalize.

- Sample: What if our sample was different? We relied on participants in the COBE Lab (mainly students, aged 19 to 25). Maybe high school students, unemployed people, or people in various industries would behave and perform differently? Maybe some would react more to a higher incentive or find the low incentive too small. The outcome could have been different, and any result might thus be an interaction (a specific combination) of the sample and the treatment. Still, we could argue that the result is likely to be similar in other samples that also consist of young people in countries where people on average are financially reasonably stable. Had our sample been even more (or less) diverse, then this would have impacted our ability to generalize.

- Setting: Setting refers to the overall environment and tasks that participants are facing in a RCT. First of all, note that we used two different tasks. If we had only used one of the tasks, our ability to generalize would have been (even) more limited. Still, what if we had chosen a different task, e.g. one that required hours to complete, rather than 20 minutes? We can't know for sure if the results hold if people were to write an essay, generate ideas, collaborate in groups etc. Again, we say that the results might be an interaction of the sample and the treatment. We can provide support for a theory, and we can have theoretical reasons for why we think the results generalize e.g. to all simple tasks that require cognitive effort. It is still a limitation though. And if we had included a very different task, our ability to generalize would have been improved. Finally, one could also consider that the task was done with paper and pencil, and one would therefore also have to consider if results can be transferred to a computer setting.
- History: What if this experiment had been run at a different time in history? In fact, some (not all) of the data was collected during the Covid-19 pandemic (Summer 2020), and we can show that the performance variation is similar both pre- and in-the-middle of Covid-19. Generally, if one has only collected data in a particular historic time (which one usually has), one cannot be absolutely certain that the particular historical context doesn't matter (the result might be an interaction of history and treatment). One then has to use theory to make sense of this limitation.
- Hawthorne: We must also reflect on the Hawthorne Effect. In any lab experiment, participants might change their behavior because they know they're being investigated. Could this awareness cause them to perform better, even for minimal incentives, than they would in less observed circumstances? Maybe one wouldn't care about 1 DKK per. task outside a context, where one has signed up to do an experiment, and potentially feels morally obligated to put in a reasonable effort.

Note that these are criteria to assess if and to what degree one can generalize findings. For example, can one expect that these findings would replicate if someone else were to run a study on the same hypothesis (do higher incentive shape performance), but using a different task, sample etc.? So, one shouldn't just state (in an exam situation for example): no, can't generalize, or yes, can generalize a little bit. Rather, one should explain in detail, using these criteria, if and to what degree a particular finding is likely to generalize to other contexts. Hence, one can argue that something might not generalize to everyone and everywhere and everything in the world, but it might generalize to situations that have XYZ features (e.g. simple tasks that require cognitive effort, while not necessarily claiming it generalizes to more complex tasks). Furthermore, consider that external validity assumes some internal validity. If internal validity had been non-existent, it would not be relevant to consider external validity. In other words, external validity can only transfer the strengths of the existing internal validity.

Finally, these are general criteria. In any particular context (e.g. doing a RCT in a project) one is also expected to rely on detailed theoretical insight. E.g., maybe literature has shown that particular kinds of tasks are similar or different, or maybe we know that old people are different from young people in this particular theoretical context. In a methods course like this, we can't expect people to be theory experts, hence we only focus on the general criteria.

Illustration of tasks used in the RCT:

8.39	6.81	8.58		8.39	6.81	8.58
3.65	9.68	3.33		3.65	**9.68**	3.33
5.57	5.55	7.05		5.57	5.55	7.05
0.32	4.54	1.09		**0.32**	4.54	1.09

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Fig 1. Math addition task.

Example of a math addition task. The left side shows what participants saw, while the right side reveals the correct solution.

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Letters	Word
osoemne	
rmbemee	
telsetr	

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Fig 2. Anagram task.

Example of tasks. Note the third task has multiple solutions. Participants only had to find one and were not rewarded for finding more than one. No participants submitted more than one answer to any of the anagram tasks.

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